Documentation

The development of Artificial Intelligence (AI), particularly Generative AI, Natural Language Processing (NLP), Recommendation Systems, and Machine Learning, has brought significant changes to various industrial sectors, including e-commerce. These technological advancements open up new opportunities to improve service quality, business process efficiency, and the user experience in digital shopping.

The AI-Powered Shopping Assistant is one innovation that utilizes this technology to help users find products that match their needs and preferences. Unlike conventional search systems that rely solely on keywords, the AI Shopping Assistant is able to understand the intent and context of user queries through more natural and interactive conversations. With NLP support, the system can interpret user needs more deeply, even when users don't know the detailed specifications of the desired product.

Furthermore, Generative AI technology enables the system to provide more informative and easy-to-understand explanations, product summaries, and recommendations. The system can analyze various data sources, such as product catalogs, product descriptions, search history, purchase history, user preferences, customer reviews, product ratings, market trends, and other user behavior data. The results of this analysis are used to generate product recommendations that are more personalized, relevant, and tailored to each user's needs.

The implementation of an AI-based Recommendation System can also assist users in the decision-making process. The system can compare multiple products based on price, specifications, quality, features, popularity, and customer reviews, then automatically present the comparison results. This eliminates the need for users to spend significant time manually searching and analyzing products. This process not only improves user convenience but also helps reduce the risk of making mistakes when selecting products.

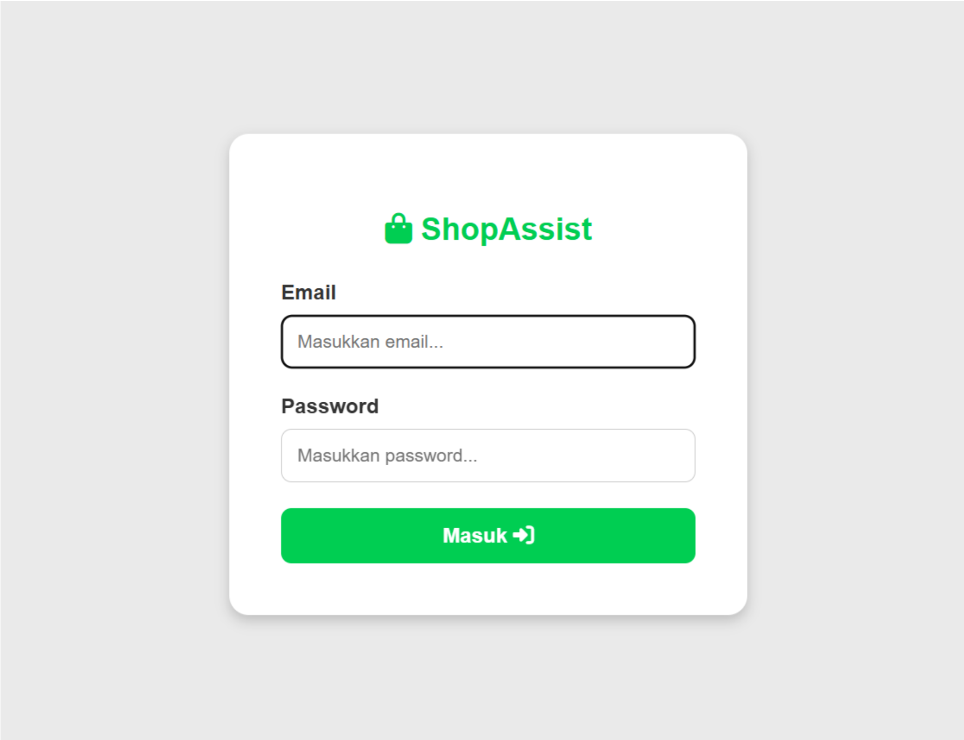
Furthermore, Machine Learning technology enables the system to continuously learn from user interactions. The more frequently the system is used, the better it understands user behavior patterns and preferences. This makes the recommendations increasingly accurate over time. The system can also dynamically adjust recommendations based on changing needs and emerging trends.

In the context of modern e-commerce, integrating AI into the product search, recommendation, and comparison process is a crucial factor in increasing customer satisfaction. Today's users want a fast, easy, personalized, and efficient shopping experience. Therefore, the development of an AI-Powered Smart Shopping Assistant & Product Decision Engine is a potential solution to address these challenges. This system functions not only as a product search tool but also as a digital assistant capable of providing intelligent recommendations, automatically analyzing product comparisons, and helping users make more informed purchasing decisions based on available data and information.

By utilizing a combination of Generative AI, NLP, Recommendation Systems, and Machine Learning, digital shopping platforms can deliver a more interactive, personalized, and user-focused experience. The implementation of this technology is expected to improve the effectiveness of the product search process, accelerate purchasing decision-making, increase customer satisfaction, and provide significant added value to the future development of the e-commerce industry.

Email and Password

Email and password are used to identify users for security purposes. Use the email and password you registered with. If they don't match, you won't be able to log in.



Product Data Acquisition and Preprocessing

Product Data Acquisition is the systematic collection of information (attributes, specifications, prices, inventory) about products from various sources, such as supplier feeds and APIs. Preprocessing transforms the often-chaotic raw data into a clean, structured dataset, ready for use by e-commerce platforms, machine learning, or analytics.

1. **Data Acquisition Methods**
   Web Scraping and Crawling is used to extract product details directly price and availability from websites.

2. **Data Preprocessing Stages**
   Data Cleaning is used to handle missing attributes for example, inferring missing weights, removing duplicates, and standardizing categorical values.

3. **Key Objectives**
   The goal will be to prepare data sets for search engine indexing or AI recommendations.

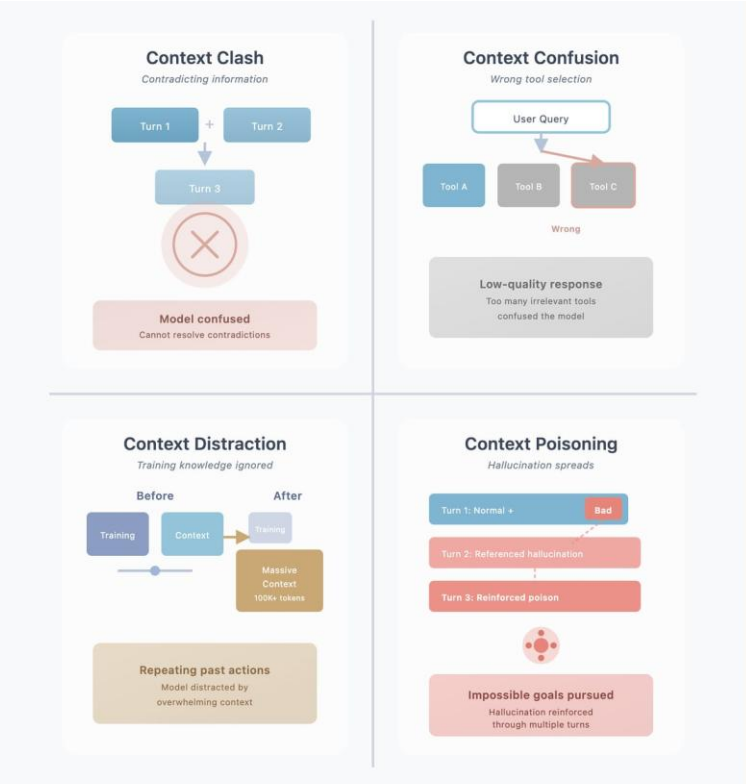
Structured Prompt Design for Product Comparison

Structured Prompt is a JSON Schema Template Prompt (a local product) created without using web scraping. Prompt engineering is the art and science of composing instructions (text) sent to an AI model so that the model responds with the exact accuracy, format, and style desired by the application. Structured Prompts are used to construct product comparison questions for Large Language Models (LLMs), which require a shift from vague, conversational questions to a specific, targeted framework. Structured prompt design uses a specific architectural framework to guide LLMs in generating consistent, accurate, and actionable product comparison content. Structured prompts for product comparisons use predefined constraints to force the AI to perform objective and in-depth analysis, detailing roles, context, and formatting, while avoiding generic summaries to produce accurate, high-quality comparison matrices and content.

Structured prompts eliminate AI ambiguity by enforcing exact formatting and objective criteria. For product comparisons, a winning prompt relies on the Role-Task-Context-Constraints-Output (RTCCO) framework: define the persona, detail the products, dictate required criteria, and constrain the output to a structured markdown matrix.

Context Pruning for Long Product Descriptions

Context Pruning for Long Product Descriptions is an artificial intelligence (AI) technique for trimming repetitive, irrelevant, or low-priority information from product documents before sending them to a Large Language Model (LLM). This dramatically improves e-commerce accuracy, cuts token costs, and reduces application response latency.



A token is the smallest unit that can be processed in an LLM model. The model can understand the concept that it inputs tokens, not words. This is because context is used to construct sentences. Tokenization is the process of replacing sensitive data with a digital representation or unique code (token). This technology is used to safeguard data processing in artificial intelligence (AI) and digitize asset ownership.

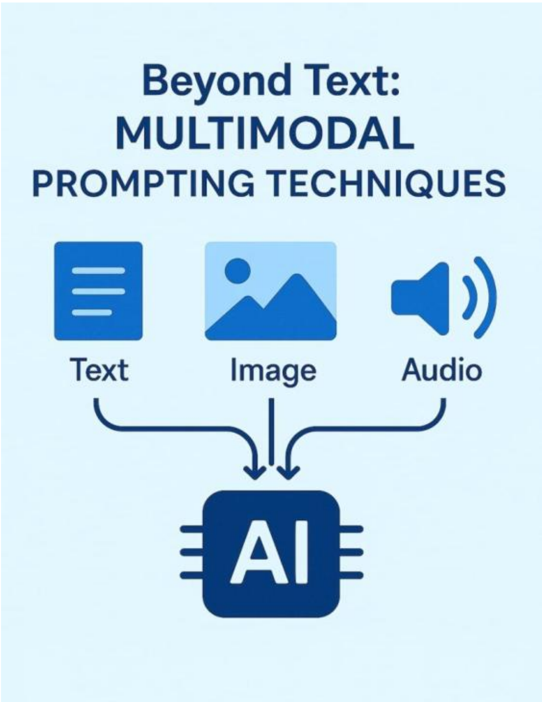
- **Text processing:** In AI like ChatGPT or Gemini, tokenization is the process of breaking down long sentences or texts into smaller pieces of words or syllables called tokens, so that they can be read and processed by machines.
- **Data analysis:** Helps computer systems understand human language patterns, translate them, and generate responses efficiently.

If an LLM model were to understand a single sentence when fed a full sentence, it would require a significant amount of memory. Because the system would use more memory, and ultimately, the more memory it uses, the greater its resources and the longer its computation time.

Context in an AI prompt is all the information the model inputs into an agent before generating a response. Context helps the model understand the prompt, and the model will respond by providing an answer based on the context inputted into the model.

Multimodal Prompting for Image-Based Product Understanding

Multimodality is important because it is a system that combines text and images. Images can help AI models understand visual data, or visual information can be used as a reference, making it easier for AI models to interpret complex scenarios that require additional information such as tables, graphs, image-based prompts, and visual prompts. This will improve the response. If multimodality functions rationally and analytically, multimodality will emerge if the model processes image-based content and visuals by integrating text-based instructions written in the prompt. This allows the text to be combined with additional image-based information to produce a response. If an AI model reads an image, the input and visual input must provide clear instructions so the AI model can understand the image and the information contained within it. The best visual-based prompts must include the context of the image being represented, for example, by inputting an image and writing it in the prompt. The output can be structured in the form of a point-based table or a list. For example, what objects are detected in the image, or in table form.



Conversational Shopping Assistant with Multi-Turn Context

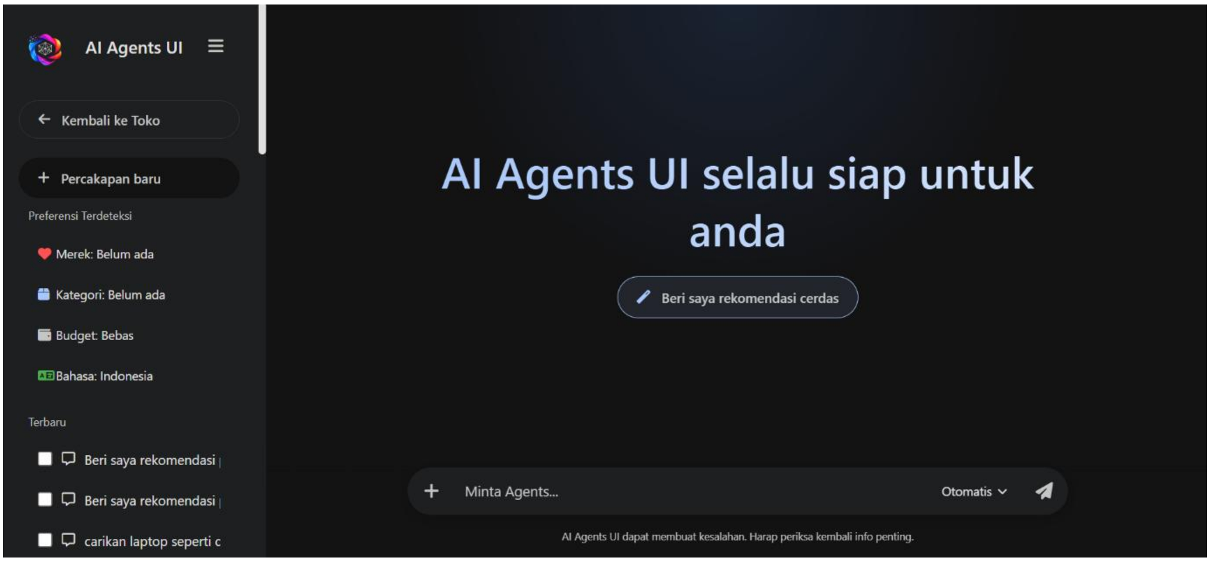
A multi-turn, context-driven conversational shopping assistant is an AI agent designed to remember previous questions and user preferences throughout a conversation. Instead of treating each request as a new interaction, it builds an ongoing dialogue to refine product recommendations, handle questions, and seamlessly complete complex shopping tasks.

Main Abilities:

- **Context Retention:** AI remembers size, color, budget, or previous items discussed in chat.
- **Dynamic Refinement:** Users can gradually narrow down choices without having to restate basic parameters.
- **Conversational Commerce:** Integrates with catalog and inventory in the backend system to provide real-time pricing and availability.
- **Multi-Step Troubleshooting Guide:** Guides users through troubleshooting product issues, sizing, or returns without the need to be redirected to a human representative.

Website

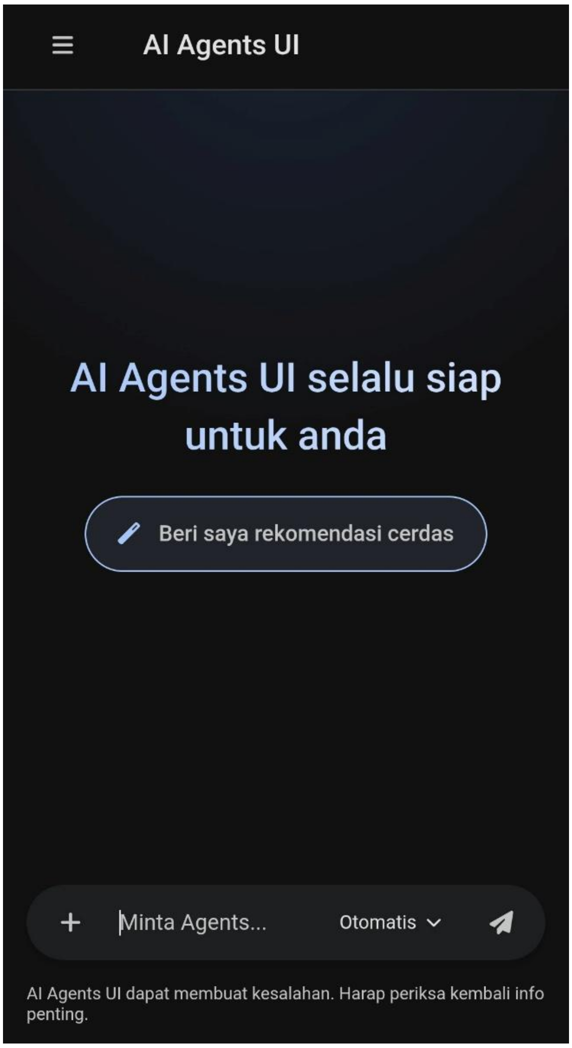
A **website** is a collection of interconnected digital pages that can be accessed via the internet using a browser, such as Google Chrome, Edge, or Safari. These pages can contain text, images, and even videos, and serve as an informational resource, portfolio, or online store. Each website typically operates under a single domain name (web address) and contains a variety of information, including text, images, videos, and interactive features.



Website Mobile

A **mobile website** is a version of a website specifically designed for easy navigation on a smartphone or tablet, typically using responsive design. Designing a site to be mobile-friendly is crucial for optimizing user experience and search engine visibility.

Mobile refers to mobile devices such as smartphones or tablets. Mobile web, or mobile web, refers to website pages specifically designed and optimized for access via mobile devices such as smartphones and tablets. With mobile internet traffic now dominating, this design focuses on loading speed, easy touch navigation, and a layout that automatically adjusts to screen size.



Mobile web applications are websites optimized for mobile devices and specifically designed to function like real mobile apps. Web-based applications are optimized to look and feel like mobile apps when opened through a mobile browser. These applications do not require installation on internal memory.

Conclusion

This system is designed to help users search, compare, and determine the best products through conversational AI and an intelligent recommendation approach. By utilizing modern artificial intelligence technologies such as semantic search, structured prompt engineering, context pruning, and multimodal AI that can simultaneously understand product text and images, the system can provide more relevant and tailored recommendations to users' needs. Users no longer have to manually search through numerous product pages, as the system can more deeply understand search intent and present the most suitable product alternatives based on their preferences, needs, and budget.

Furthermore, the system allows users to automatically compare products based on various important aspects such as price, specifications, features, quality, customer reviews, and overall product value. Through conversational interactions, users can ask questions naturally, much like discussing with a personal shopping consultant. This approach improves the user experience by making the search and decision-making process faster, easier, and more personalized.

The application of AI in e-commerce also provides significant benefits for both marketplace platforms and sellers. AI-based shopping assistants can enhance customer engagement through more responsive and personalized interactions, speed up the product search process, and help users find the most relevant products in less time. This has the potential to increase sales conversion rates, reduce cart abandonment rates, and drive increased customer satisfaction.

Furthermore, the system can also help businesses understand consumer behavior more deeply through user interaction data analysis. The information obtained can be used to identify market trends, customer preferences, purchasing patterns, and opportunities for product development and more effective marketing strategies. Therefore, the implementation of AI-Powered Smart Shopping Assistants & Product Decision Engines not only benefits consumers in making more informed purchasing decisions but also supports digital business growth through optimal use of data and artificial intelligence.